A counterexample to the higher Toeplitz Hurwitz-stability conjecture

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Task 1. Resolve the stability conjecture

Karp's Conjecture 5 [1] asserts that the rising-factorial determinant polynomial $P_n^r(x)$ is Hurwitz stable whenever the input sequence belongs to PF_r, for $r \geq 2$. Hurwitz stability means that every zero has strictly negative real part. The order-two specialization, explicitly included in the source, is

$$P_n^2(x) = \sum_{k=0}^{n} \binom{n}{k} f_k f_{n-k} \big[(x)_k (x)_{n-k} - (x+1)_k (x-1)_{n-k}\big], \tag{1}$$

where $(x)_k$ is the rising factorial.

Theorem 1. *Take $r = 2$, $n = 15$ and*

$$(f_0, \ldots, f_{15}) = (1, 51, 52, 53, 54, 55, 56, 57, 57, 56, 55, 54, 53, 52, 51, 1).$$

This is a positive strictly log-concave PF_2 sequence, but P_{15}^2 has exactly one zero in

$$\left| x - \frac{1 + 41i}{10} \right| < \frac{1}{20}.$$

In particular it has a zero with $\operatorname{Re} x > 1/20$, and Conjecture 5 is false.

The input satisfies the full hypothesis

The adjacent log-concavity differences are

$$f_k^2 - f_{k-1} f_{k+1} = \begin{cases} 2549, & k = 1, 14, \\ 57, & k = 7, 8, \\ 1, & k = 2, \ldots, 6,\ 9, \ldots, 13. \end{cases}$$

Thus strict log-concavity holds at every required index.

For completeness, extend f_k by zero outside $0 \leq k \leq 15$. All entries of its infinite Toeplitz matrix are nonnegative. A two-by-two minor has the form

$$f_a f_{a+h-d} - f_{a+h} f_{a-d}, \qquad h, d > 0$$

for integer indices. If the second product is zero, the minor is nonnegative. Otherwise all indices from $a - d$ through $a + h$ are in the positive support. Since the ratios f_j / f_{j-1} decrease,

$$\frac{f_a}{f_{a-d}} \geq \frac{f_{a+h}}{f_{a+h-d}},$$

by multiplying d successive ratio comparisons. The minor is again nonnegative. This proves PF_2 directly, including every minor of the infinite matrix.

An explicit polynomial factorization

Expanding the finite sum (1) gives

$$P_{15}^2(x) = 210(x+1)_7 R(x), \tag{2}$$

where

$$R(x) = 191647x^6 + 4104969x^5 + 36750955x^4 + 192645735x^3 + 736979998x^2 + 2128319736x + 3212352000.$$

This is an integer polynomial identity. It can be checked by ordinary coefficient multiplication. The accompanying verifier also checks it by evaluating both sides independently at $x = 0, 1, \ldots, 15$; both sides have degree at most 15, so these sixteen exact evaluations alone certify the identity.

All coefficients of the displayed factorization are positive. Thus the counterexample concerns stability; it does not contradict coefficient positivity.

An integer certificate for a right-half-plane zero

Put

$$T(w) = 20^6 R\left(\frac{2 + 82i + w}{20}\right) = \sum_{j=0}^{6} c_j w^j.$$

Its coefficients are the following Gaussian integers:

j	$\operatorname{Re} c_j$	$\operatorname{Im} c_j$
0	6242984901271040	-4815010659523584
1	-7146943743914496	-9080441566052096
2	-369163407187520	-56477312908800
3	-4012914867840	2981422413440
4	-3796641800	34603649040
5	84399144	94290324
6	191647	0

Let

$$L = 11555657658184807, \qquad U = \sum_{j \neq 1} \left\lceil \sqrt{(\operatorname{Re} c_j)^2 + (\operatorname{Im} c_j)^2} \right\rceil = 8262602600766508.$$

Direct integer squaring verifies $L^2 \leq |c_1|^2$, and integer square root bounds give the displayed exact value of U. Moreover,

$$L - U = 3293055057418299 > 0.$$

Consequently on $|w| = 1$,

$$|T(w) - c_1 w| \leq \sum_{j \neq 1} |c_j| \leq U < L \leq |c_1 w|.$$

Rouché's theorem implies that T and $c_1 w$ have the same number of zeros inside the unit disk, namely one, counted with multiplicity. Undoing the affine change of variable gives exactly one zero of R in the claimed disk. The factor $(x+1)_7$ has its zeros at $-1, -2, \ldots, -7$, outside this disk, so the same statement holds for P_{15}^2. Every point in the disk has real part greater than $1/20$. This completes the refutation.

Verification and scope

The verifier uses only exact integer arithmetic and Python's standard library. It checks the input inequalities, constructs the original determinant polynomial, checks (2) by coefficients and by independent evaluations, and computes T by both binomial expansion and Horner substitution. The norm comparisons use integer square roots. No approximate root finder is used in the certificate.

The result refutes the universal assertion through its allowed $r = 2$ case. It does not decide the separate assertions for each $r \geq 3$, or the stronger PF_∞ real-root conjecture. The source's previously known nonreal-zero examples under mere log-concavity do not assert failure of Hurwitz stability. The published statement, live MathDB status and public claims were checked on 11 September 2026; no matching existing stability-refutation claim was located. Details are in `literature-audit.md`.

Tasks 2–5

2. Determine the smallest degree index n for which positive strictly log-concave data give an unstable order-two determinant polynomial.

3. Determine the instability ranges in A and n for $f_0 = f_n = 1$ and $f_k = A + \min(k, n-k)$ at the interior indices.

4. Prove or refute Hurwitz stability separately for each determinant order $r \geq 3$ under the corresponding PF_r hypothesis.

5. Resolve the proposed real-negative-zero property under the stronger PF_∞ hypothesis.

References

[1] D. Karp, *Positivity of Toeplitz determinants formed by rising factorial series and properties of related polynomials*, Zap. Nauchn. Sem. POMI **404** (2012), 184–198, Conjecture 5 and the order-two identity, p. 193. Published article; arXiv:1203.1482. English translation: Journal of Mathematical Sciences **193** (2013), 106–114, doi:10.1007/s10958-013-1438-y.